

De-Americanization

Removing the early-exercise premium from listed option quotes, fast

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Abstract

Chains whose snapshot declares `exercise_style = "american"` — in practice US-listed single-stock and ETF options — may be exercised early, so their prices are worth *at least* their European counterparts; the excess is the early-exercise premium (EEP), and it may be zero. Feeding such prices straight into a European implied-volatility inversion biases the smile upward, worst where vega is small. This note documents the Vol-Fitter’s de-Americanization: for each quote it finds the volatility σ^* at which a Cox–Ross–Rubinstein American tree reprices the quote’s *mid*, subtracts the implied premium $\widehat{\text{EEP}} = \max(A^{\text{mid}} - P_B(\sigma^*), 0)$ from bid, mid and ask alike (one root per strike, dollar spread preserved), and hands the shifted band to the European pipeline. We place the step precisely in the production quote pipeline, explain why the tree inversion is the right control variate, how the tree’s carry is chosen (a forward-consistent discrete cash-dividend schedule, or a continuous-carry fallback), and the numerical choices — tree depth, the bracketed-bisection sweep — that hold the inversion error budget to a few vol bps. The hot path is a Numba kernel: on a 300-quote synthetic chain it runs in 11.56 ms versus 998.9 ms for the NumPy fallback through the *same* dispatch and bracketing (86×; both paths return identical roots), and a prepared-quote content cache ensures fit tuning never re-runs the trees. Because the inversion is per strike, the European-equivalent call curve can come out slightly non-convex in the sparse wings; an approximate, band-constrained *wing* projection repairs it while leaving the at-the-money core byte-identical — the first, global repair moved the core and was reverted. On a synthetic 6%-dividend name the naive inversion overstates IV by up to 282 vol bps; on a captured SPY chain the same wedge runs to a median 519 vol bps — but only on deep in-the-money puts the fitter *discards*: on the quotes production actually fits, the de-Americanization moves the implied vol by a median $|\cdot|$ of 4.2 vol bps (maximum 60).

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1 Why de-Americanize, and for which chains

The volatility models of Notes 01–04 are *European*: they model the normalized undiscounted call

$$c(k) = \mathbb{E}^T[(e^{X_T} - e^k)^+], \quad X_T = \log(S_T/F), \quad (1)$$

a terminal expectation under the T -forward measure. A listed American option is worth *at least* that: its holder may exercise before expiry, and the excess — the *early-exercise premium* (EEP) — is strictly positive exactly when early exercise carries value (puts under positive rates, calls around dividends) and zero otherwise. If one inverts an American price with the European Black formula, whatever premium is present is misattributed to volatility, and the implied vol comes out biased upward.

The machinery of this note engages only when a chain snapshot declares `exercise_style = "american"` (set by the data layer; in this codebase that means US-listed single-stock and ETF chains). European snapshots — index options, synthetic chains — skip every step below byte-identically.

Conventions. Throughout: S is spot, F the forward and D the discount factor to expiry (both resolved from the chain itself, Note 06), K a strike, $k = \log(K/F)$ log-moneyness, $r = -\log(D)/t$ and $q = r - \log(F/S)/t$ the continuously compounded rate and carry yield the pair (F, D) implies. t is the *calendar* year fraction to expiry and τ the event-weighted variance-clock years (Note 00); $w = \sigma^2\tau$ is total implied variance and $z = k/\sqrt{w_{\text{atm}}}$ standardized moneyness, where w_{atm} is the at-the-money total variance. A *normalized call price* is the option price in undiscounted forward units, $c = p/(DF)$ plus the parity shift of eq. (3). One *vol bp* is 10^{-4} of absolute volatility (a move from 25.00% to 25.01% is one vol bp).

Invariant

The de-Americanization machinery protects five production invariants:

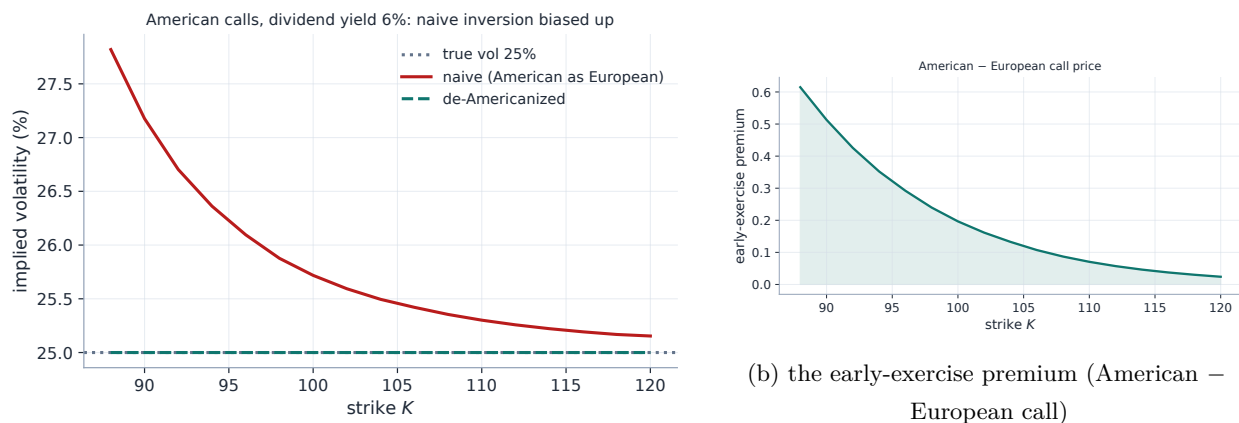
1. European chains are untouched — every step below is a byte-identical no-op unless the snapshot is American.
2. One premium per strike, subtracted from bid, mid and ask alike: the quoted *dollar spread* survives de-Americanization.
3. A failed inversion never invents an adjustment: the premium is taken as zero and the raw band falls through to the European static-bound filters.
4. The dense at-the-money core is never moved by the convexity repair (section 7).
5. Fit settings — model choice, penalties, weights, optimizer tuning — never re-run the trees (the prepared-quote cache, section 6).

Heuristic

The bias is not uniform. A price error ΔP moves implied vol by $\Delta\sigma \approx \Delta P/\text{vega}$, so a given dollar of unremoved early-exercise premium translates into a *large* spurious vol move exactly where vega is *small* — deep in the money, in the wings. (Conversely, at the money, where vega peaks, the same dollar error is attenuated.) A naive smile built from American quotes therefore

has corrupted wings, the very region the fitter most needs to extrapolate. De-Americanization is not a cosmetic preprocessing step; it is the difference between fitting the market's volatility and fitting its exercise policy.

Figure 1 makes this concrete on American calls written on a stock paying a 6% dividend yield (early exercise to capture dividends is rational): an option priced at a true 25% volatility, inverted naively as European, implies a volatility up to 282 vol bps too high, rising as the option goes in the money. De-Americanization recovers a flat 25%. Section 5 works one strike of this figure through by hand.



(a) naive vs de-Americanized implied vol

(b) the early-exercise premium (American - European call)

Figure 1: The naive European inversion of an American price is biased up by the early-exercise premium; the de-Americanization returns the true volatility. Synthetic American calls, 6% dividend yield, true vol 25%.

2 Where the step sits: the production quote pipeline

De-Americanization is one stage of quote preparation (`volfit/api/quotes.py`). The full flow for one (ticker, expiry) is

```
raw chain → resolve  $F, D$  → OTM / tick screens → root on mid → shift bid/mid/ask →
normalized call → total variance → optional convex repair → model.
```

In words: the forward and discount are resolved first (put-call parity plus the American de-bias of section 3.2); the OTM side of the book is kept — puts for $K < F$, calls for $K \geq F$, because the OTM option carries the vega and the tight relative spread — and tick-noise / missing-side screens run; then, on American chains only, each surviving *mid* is de-Americanized. The tick screen deserves its one sentence: on real-feed chains an OTM quote whose *bid* is worth no more than three ticks is dropped — a couple of ticks carries no volatility information and whipsawed the wings — with the tick size persisted alongside the snapshot so replays screen identically (certification case `tick_noise_quotes`).

2.1 One root per strike, spread preserved

Given the American mid A^{mid} at a strike, the pipeline solves for the volatility σ^* at which the CRR American tree of section 3 reprices it, then reprices the *European* leg at that same volatility:

$$A_{\text{CRR}}(\sigma^*) = A^{\text{mid}}, \quad E^{\text{mid}} = P_B(\sigma^*), \quad \widehat{\text{EEP}} = \max(A^{\text{mid}} - E^{\text{mid}}, 0). \quad (2)$$

The clamp at zero exists because the premium is theoretically nonnegative while quote noise (and tree error) can dip the estimate below; it is one-sided by design, and section 7 returns to its cost. The *same* $\widehat{\text{EEP}}$ is subtracted from bid, mid and ask, so the quoted dollar spread is preserved at the cost of one root find per strike instead of three. Quotes the tree cannot invert keep $\widehat{\text{EEP}} = 0$ and their raw prices fall through to the European static-bounds filter (section 4.2).

2.2 Into model space: parity, variance, clocks

After the shift, the adjusted prices p enter the ordinary European pipeline: normalization to undiscounted forward units and put–call parity conversion of the (OTM) puts to normalized calls,

$$c = \frac{p}{DF} \quad (\text{calls}), \quad c = \frac{p}{DF} + 1 - e^k \quad (\text{puts}), \quad (3)$$

then a strike-by-strike inversion to total implied variance w . Strikes whose bid, mid or ask violates the static bounds $(1 - e^k)^+ < c < 1$ are dropped (a one-sided band cannot be weighted), and a wing filter removes strikes beyond $Z_{\text{max}} = 4$ ATM standard deviations.

One clock subtlety: the total variance w is inverted from the *price*, so it is clock-independent; only the *reported* implied vol depends on the working clock, $\text{iv} = \sqrt{w/\tau}$. The σ^* of eq. (2) is a calendar-time Black vol used transiently inside the premium estimate; the vol a user sees on screen for the same quote can differ because $\tau \neq t$ when event weighting is on (Note 00).

3 The pricing tree

3.1 Cox–Ross–Rubinstein

The American value is computed by a Cox–Ross–Rubinstein binomial tree [1]. With N steps of size $\Delta t = t/N$, the spot moves by $u = e^{\sigma\sqrt{\Delta t}}$ or $d = 1/u$ each step, with risk-neutral up-probability

$$p = \frac{e^{(r-q)\Delta t} - d}{u - d}, \quad 0 < p < 1, \quad (4)$$

(r the rate, q a continuous carry yield). The condition $0 < p < 1$ is the CRR no-arbitrage requirement $d < e^{(r-q)\Delta t} < u$; it fails only for extreme drift-to-vol ratios, and the scalar path then *raises* rather than silently mis-pricing (the batch path differs; section 4.2). The tree rolls back from the payoff with discounting $e^{-r\Delta t}$, taking $\max(\text{continuation}, \text{intrinsic})$ at each node for the American feature. The European value is the same recursion without the intrinsic comparison. Stripped of the escrowed-dividend machinery, the kernel is a dozen lines (this reproduces `binomial_price` at $q = 0$ to 10^{-10} ; the European leg matches Black–Scholes to 2×10^{-3} at depth):

Listing 1: The CRR backward induction with early exercise (distilled from `core/american.py`).

```
1 def crr_price(is_call, s, k, t, sigma, r, n=501, american=True):
```

```

2     dt = t/n; u = np.exp(sigma*np.sqrt(dt)); d = 1/u
3     p = (np.exp(r*dt) - d)/(u - d); disc = np.exp(-r*dt)    # risk-
        neutral up-prob
4     sT = s * u**np.arange(-n, n+1, 2)                        # terminal
        spots
5     v = np.maximum(sT-k, 0.0) if is_call else np.maximum(k-sT, 0.0)
6     for m in range(n-1, -1, -1):                            # roll back
        one layer
7         v = disc*(p*v[1:] + (1-p)*v[:-1])                    #
        continuation value
8         if american:                                        # early-
            exercise test:
9             sM = s * u**np.arange(-m, m+1, 2)                # max(
                continuation,
10            v = np.maximum(v, sM-k if is_call else k-sM)      #
                intrinsic)
11     return v[0]

```

3.2 Carry and dividends: what the tree actually uses

For a real single name a flat continuous yield is a crude proxy: the call-side early-exercise decision keys on discrete ex-dates. Production therefore chooses, per (ticker, expiry), *one* of two carry models for the tree (volfit/api/quotes.py, volfit/data/dividends.py):

- **Discrete cash schedule**, $q = 0$. When the dividend model supplies cash dividends with ex-dates in $(0, t]$, the tree prices them by the *escrowed-spot* method [2]: the present value of future dividends is stripped from the spot to form the diffusing base lattice, and the actual spot at each node is the base plus the PV of dividends not yet paid — recombining (cash drops otherwise break recombination) while exercising against the true cum-dividend spot. The cash amounts are *rescaled* by

$$\alpha = \frac{S - Fe^{-rt}}{\sum_i D_i e^{-r\tau_i}} \quad (5)$$

so the escrowed-tree forward $(S - PV_0)e^{rt}$ reproduces the resolved forward *exactly*: the timing stays the forecast, the amounts bend to the market. A non-physical α (≤ 0 or absurdly large) rejects the schedule.

- **Continuous carry fallback**. Otherwise the tree runs on the (r, q) implied by the resolved pair (F, D) : $r = -\log(D)/t$, $q = r - \log(F/S)/t$.

The two never coexist inside one tree: it is cash-with- $q=0$ or continuous-yield, and eq. (5) is what keeps the cash choice consistent with a forward whose level partly reflects yield-like far-tail effects.

Remark 1 (The apparent circularity, resolved). De-Americanization needs (F, D) ; but put–call parity — the source of (F, D) — is an *equality only for European options*, so raw American mids bias the parity regression. Production resolves this in order: the parity regression runs first, then `_refine_american` (volfit/data/forwards.py) de-biases it with a *coarse* de-Am pass — including, where the de-Americanized put and call IVs fail to join at the money, a rate bisection that zeroes

that gap (the joint (r, F) refinement; the ATM-kink case file of Note 06). Only then does the *precise* per-quote de-Americanization of this note run, under the refined carry. The refinement uses fewer bisections and a coarser tolerance than quote prep — it needs the carry to a few bps, not the quote to sub-bp. One step deeper exists as an explicit opt-in: the *joint borrow/de-Am fixed point* of Note 06 (`jointCarry`) iterates the full loop — borrow and dividends into de-Am, de-Am into the parity forward, the forward back into borrow — to convergence, and routes the converged (F, D) into the fits per expiry *only* when the implied borrow clears a materiality gate; below it the parity forward is kept exactly, so ordinary names are byte-identical even with the toggle on.

4 The root find: control variate and error budget

4.1 Tree inversion as a control variate

De-Americanization is a one-dimensional root find. Given an American quote A at (S, K, t, r, q) ,

$$\boxed{\text{find } \sigma^* : \text{CRR}_{\text{Am}}(\sigma^*) = A, \quad \text{return } \sigma^* \text{ as the European-equivalent Black vol.}} \quad (6)$$

This is the standard desk workflow — convert American quotes to pseudo-European quotes, then calibrate a European model — studied critically by Burkovska et al. [5]: pragmatic and fast, not theoretically exact outside the model.

Heuristic

Why is the *tree's* σ^* the right European volatility? Because the tree is a *control variate* for the early-exercise feature. The market price differs from the European price we want by an EEP we cannot observe; the tree supplies a model EEP as a function of σ . Subtracting it at σ^* ,

$$A_{\text{mkt}} - \underbrace{[A_{\text{CRR}}(\sigma^*) - E_{\text{CRR}}(\sigma^*)]}_{\text{model EEP at } \sigma^*} = E_{\text{CRR}}(\sigma^*) \approx P_B(\sigma^*), \quad (7)$$

where the first equality uses the defining property $A_{\text{CRR}}(\sigma^*) = A_{\text{mkt}}$ and the last step is tree discretization only ($\sim 10^{-5}$ of spot at the scalar depth). So σ^* is the de-Americanized Black vol: the early-exercise premium cancels in the control variate, and the tree's *level* error largely cancels with it — the tree only has to price the American/European *difference* well. The method is exact in the model and robust in practice because the EEP is a smooth, slowly varying function of σ .

4.2 Static bounds, bracketing, and failure semantics

A price below intrinsic (impossible for an American option) or above the static upper bound (S for a call, K for a put) yields `nan`, mirroring the fitter's convention for unusable quotes. The bracket lower end is lifted just above the CRR drift floor $\sigma\sqrt{\Delta t} > |r - q|\Delta t$ so eq. (4) stays valid, and the upper end is expanded until the price is bracketed, capped at $\sigma = \text{SIGMA_HI} = 4$.

The scalar and batch paths deliberately differ in how they fail. The *scalar* path (one quote, Brent root find, $N = 501$) *raises* on an invalid CRR probability — a single mis-priced input should be loud. The *batch* path must survive mixed inputs, so invalid lanes (static-bound violations, a deep-ITM put priced at its early-exercise floor below the drift-floor model price, never-bracketed quotes) return `NaN` and the rest of the chain proceeds. Upstream, quote prep treats a `NaN` root as

“no premium estimate”: $\widehat{\text{EEP}} = 0$, and the untouched band is left for the European static-bound filters to keep or drop.

4.3 Depth, bisection, and the error budget

Two depths are used. The scalar path uses $N = \text{DEFAULT_STEPS} = 501$; the batch path a coarser $N = 192$ tree and a fixed 24-sweep bracketed bisection. The bisection count was tuned down from 45 to 24 at no measurable accuracy cost (< 0.01 vol bp drift, test-locked), a $\sim 1.8\times$ saving on the isolated de-Am rail.

Separating the error sources of the returned σ^* :

- **Root tolerance.** 24 halvings of a ≤ 4.0 bracket leave $\sim 2.4 \times 10^{-7}$ of σ (~ 0.002 vol bp) — pure bisection uncertainty, negligible.
- **Finite tree.** The 192-step European leg prices within a few 10^{-4} of Black; through vega that is a few vol bps on liquid quotes — visible as the ± 2 vol bp residuals on the real-chain OTM calls of section 8 (where the true EEP is ≈ 0 and the tree error is all that remains). The scalar 501-step leg is $\sim 10\times$ tighter.
- **Carry / dividend model.** A wrong forward, discount or cash schedule biases σ^* directly; de-Americanization inherits, and cannot repair, the inputs of section 3.2 (see the closing caution of section 8).
- **Market vs model.** The CRR diffusion between ex-dates and the one-sided clamp of eq. (2) are modelling choices; their footprint is the wing non-convexity that section 7 repairs.

Caution

The tree depth is a numerical-target knob, not a free speed dial. Cutting the batch depth from 192 to 128 is faster (the backward induction is $O(N^2)$, so $\sim 2\times$) but shifts the priced European leg by ~ 15 vol bps — i.e. it changes the *answer*, not just the runtime. The depth is held at 192 and speed is bought elsewhere (the Numba kernel, section 6.2).

5 A worked example, by the numbers

Case file: one strike of fig. 1

Setup: American call, $S = 100$, $K = 90$, $t = 0.5$, $r = 2\%$, $q = 6\%$, true volatility $\sigma = 25\%$, scalar tree depth $N = 501$. The production tree prices

$$A_{\text{CRR}} = 11.7856, \quad E_{\text{CRR}} = 11.2725, \quad \widehat{\text{EEP}} = A_{\text{CRR}} - E_{\text{CRR}} = 0.5131.$$

With a 6% yield against a 2% rate, exercising early to capture the dividend stream is valuable on an ITM call — about 4.4% of this option’s value. Handing the American price to the analytic European Black inversion (production’s normalization: $c = A/(DF)$ at $k = \log(K/F)$) returns

$$\sigma_{\text{naive}} = 27.19\%,$$

a +219 vol bp phantom: the inversion attributes the premium to volatility. Solving eq. (6) instead gives $\sigma^* = 25.00\%$ — the truth, to bisection tolerance. Every number above is generated by `figures/gen_deam.py` from `core/american.py`; the full-strike sweep is fig. 1.

6 Caching and the compiled kernel

6.1 The prepared-quote cache key

De-Americanized, inverted quotes depend only on the *data* entering quote preparation — never on any fit setting. The pipeline memoizes each node’s prepared quotes on a content-digest key (`volfit/api/service.py`) built from:

```
(ticker, expiry, data_version, forward, discount, cash-dividend digest,  $t$ ,  $\tau$ , as-of)
```

`data_version` is the raw chain’s identity (bumped on fetch or chain invalidation); the forward/discount absorb the forward policy and manual edits; the cash digest absorbs the dividend model and rate; calendar t drives the CRR carry and discounting; τ is included because the *cached reported IV band* is quoted in the event-weighted clock ($iv = \sqrt{w/\tau}$), so an event edit must re-key. A change to any of these re-runs the trees; model choice, penalties, quote weights and optimizer tuning do *not* — so the de-Am cost is paid once per genuine data change and is then free across an entire calibration sweep. A conservative pre-screen (non-positive bids, a buffered wing cut) additionally drops quotes that cannot survive the static bounds *before* any tree runs, and is output-preserving by construction.

6.2 The Numba kernel

The hot path de-Americanizes a whole chain. The NumPy batch vectorizes across quotes but allocates (n_{quotes}, m) slabs per time step; the compiled kernel does a single scalar CRR per quote with no per-step allocation — precomputed power tables, the escrow lattice constants hoisted out, and a `prange` parallel loop over quotes.

Performance

On a 300-quote synthetic chain the kernel runs in 11.56 ms versus 998.9 ms for the NumPy fallback — 86 $\times$. The comparison is like-for-like: *both* timings call the public `deamericanize_batch` dispatch (identical static screens, identical drift-floor bracket), the NumPy leg by forcing the fallback; both paths run once untimed first (JIT compile excluded) and the fastest of five repeats is reported; the generator *raises* unless the two paths return identical finite/NaN lanes (206 of 300 here — the rest are deep-ITM puts at their exercise floor, correctly NaN on both) and identical roots (observed max $|\Delta\sigma| = 0$). Measured on a 20-thread desktop; this uniform synthetic chain parallelizes especially well — the documented figure on wide *real* chains is $\sim 60\times$. If Numba is unavailable the fallback serves with identical results; the kernel agrees with the scalar tree to tree rounding.

7 Cross-strike convexity: repairing the de-Americanized wings

De-Americanization inverts each strike *independently* — its own root find (6), its own one-sided clamp (2) — with no coupling across strikes. That independence is what makes the kernel embarrassingly parallel (section 6.2), but the European-equivalent call prices $\{c(K_i)\}$ it hands the parity conversion (3) need not come out *convex* in the strike. To be precise about the mechanism: independent inversion does not by itself create arbitrage — if the input prices were exactly consistent with one arbitrage-free model, the outputs would be too. The non-convexity is injected by quote noise in sparse wings, the one-sided $\max(\text{EEP}, 0)$ clamp (a strike-local, asymmetric perturbation), finite-tree

error, and carry/model mismatch — and per-strike inversion simply has no cross-strike coupling with which to absorb any of it.

Convexity is not optional. Writing the undiscounted call as $C(K) = \mathbb{E}^\mathbb{Q}[(S_T - K)^+]$, its second derivative in strike is the risk-neutral density, $\partial^2 C / \partial K^2 = f_{S_T}(K) \geq 0$: *convexity of the call curve in strike is exactly the no-butterfly-arbitrage condition*. On a discrete strike grid the test is the divided second difference

$$\Delta_i^2 = \frac{C_{i-1}(K_{i+1} - K_i) - C_i(K_{i+1} - K_{i-1}) + C_{i+1}(K_i - K_{i-1})}{K_{i+1} - K_{i-1}} \geq 0, \quad (8)$$

non-negative iff C is convex at K_i . Because the injected noise concentrates in the sparse, low-vega wings, Δ_i^2 can come out slightly negative there — a genuine (small) butterfly arbitrage in the *input* handed to every downstream model, and the seed of the put-wing violation of the Multi-Core Sigmoid model that Note 03 dissects.

Caution

The reverted global fix. The first repair projected the *whole* call curve onto the convex cone with a free *global* affine part. Repairing a wing dip that way re-tilted the baseline and moved the at-the-money prices — prices the repair had no business touching. The ATM core is where vega peaks, so the sub-penny price moves were *small* in implied vol; but the core is also where quotes are densest, spreads tightest, and fit weight heaviest, so the unnecessary coherent shift — and the discontinuity it opened against neighbouring untouched quotes — surfaced immediately as a visible ATM smile gap on live SPY and NVDA. It was reverted. The lesson is about *locality*, not vega: a repair aimed at the wings must be structurally incapable of moving the trusted core, hence invariant 4. The revert and the confined repair’s clean-chain no-op are locked together as certification case `deam_repair_confinement`.

The shipped repair (`volfit/calib/convex_deam.py`) is therefore *wing-only and band-constrained*, and honestly: *approximate*. It acts on the normalized OTM call curve *after* the parity conversion (3). Strikes in the core $|z| \leq Z_{\text{core}} = 1$ are held byte-identical. Each wing, anchored at its core boundary, is replaced by an approximation to the ℓ^2 -closest curve that is simultaneously *convex* and *inside the quoted band* — the projection onto the intersection $\mathcal{C} \cap \mathcal{B}$ of

$$\mathcal{C} = \{\hat{C} : \hat{C} \text{ convex, leaving the anchor no flatter than the core}\}, \quad \mathcal{B} = \{\hat{C} : C^{\text{bid}} \leq \hat{C} \leq C^{\text{ask}}\}, \quad (9)$$

computed by *at most 25 iterations* of Dykstra’s alternating projection — the convex-cone step a non-negative-hinge least squares, the band step a clip. The implementation does not certify that $\mathcal{C} \cap \mathcal{B}$ is nonempty (a wing whose convexification demands leaving the band has no exact solution; the iterate then lands near the band), does not post-check global convexity of the final assembled curve, and by construction cannot repair a violation *inside* the held-fixed core. A common per-strike shift is applied to bid, mid and ask so the *spread is preserved*, and the repaired band is re-inverted to total variance.

Heuristic

Why the band constraint is load-bearing. Plain convex projection of an illiquid, genuinely non-convex wing is free to push a price all the way to the static no-arb boundary — and a wing call price at that boundary inverts to an absurd implied volatility (a put wing was seen to jump

from 27% to 104%, itself violating Lee’s moment bound $w(k) \leq 2|k|$ for large $|k|$; Note 03), detonating the downstream fit. Confining the repaired price to the *quoted spread* bounds the correction to the market’s own stated uncertainty: the repair can move a price only as far as the bid–ask allows, which is exactly as far as the quote is genuinely ambiguous.

The repair is gated American-only and short-circuits on any curve already convex to a tolerance $\text{CONVEX_TOL} = 10^{-3}$ (normalized price units), calibrated so that dense liquid chains — whose de-Am wings carry only $\sim 10^{-4}$ rounding curvature — are *never* touched (byte-identical, ATM implied-vol difference $\sim 10^{-15}$), while genuinely arbitrated illiquid wings (10^{-3} to 10^{-2}) are repaired. On those illiquid nodes an ablation found the repair both *reduces* the fitted arbitrage and *lowers* the in-sample error — the model no longer chases the arbitrated quotes — and it composes with the put-wing penalty of Note 03 (a Durrleman-style $g \geq 0$ constraint on the *output* smile) rather than duplicating it: the de-Am repair removes the arbitrage at the *source*, and running both attains the penalty’s arbitrage removal at a fraction of its standalone in-sample cost.

One boundary hardening came from the field rather than the design. Multiple listings can quote the *same strike* (weekly and monthly series overlapping), which puts two identical abscissae into the wing projection — and the core-anchor *slope*, a divided difference across the boundary, divided by zero: the crash killed whole backtest day-pairs before it was traced. The anchor slope now reads the nearest *strictly distinct* strike, clean chains are byte-identical, and the case is certification-locked (`duplicate_strikes`).

8 Real-chain evidence, honestly labelled

The synthetic example of section 5 controls the truth; figs. 2 and 3 show the effect on live market prices, with nothing simulated — and on two deliberately different populations.

The stress exhibit (fig. 2). Every two-sided put quote of one captured SPY expiry (the Massive true-weekly fixture `lv_weekly_massive.json`, as-of 2026-06-25; the December-2026 expiry, strikes 55%–138% of spot) is inverted both ways: naively through the analytic European Black formula (production’s inversion, what a fitter that ignored early exercise would compute) and through the American tree (6). Since the EEP itself has *price* units, what the wedge between the curves shows is the *model-implied IV effect* of the premium, not an observed premium. Out of the money the two agree to a few vol bps; in the money the wedge runs to a median 519 vol bps across the 54 ITM puts, up to 814.

But this population is **not what Vol-Fitter fits**. Production keeps the OTM side only — puts below F , calls above (section 2) — so the ITM puts carrying the headline wedge are, in all but the thin band between spot and the forward, exactly the quotes production *discards* in favour of their OTM call twins. Figure 2 is an economic stress illustration of what skipping de-Americanization would do to the discarded side, not a statement of fit exposure.

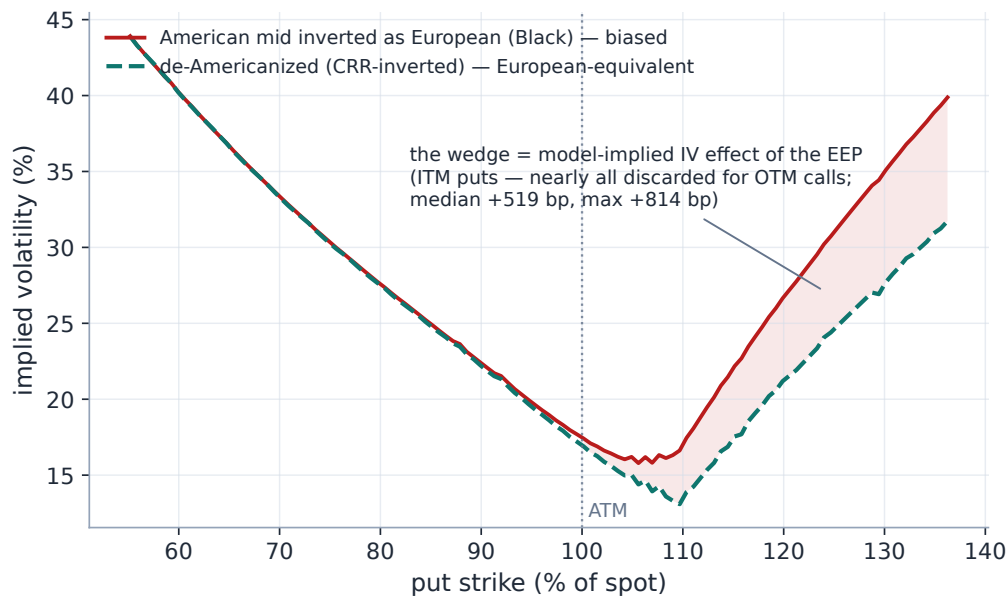


Figure 2: Real market data, one live SPY expiry (Massive capture, as-of 2026-06-25, December-2026): every two-sided put mid inverted as if European versus de-Americanized. The ITM wedge — median 519 vol bps, up to 814 — is the model-implied IV effect of the early-exercise premium on a side production almost entirely *discards* for OTM calls: a stress exhibit, not the fitted population (that is fig. 3). Carry supplied ($r = 4.3\%$, $q = 1.2\%$) because the delayed capture's parity carry is unusable (Note 06). Generated by `figures/gen_deam_real.py` from the production CRR machinery.

The fitted population (fig. 3). Running the same two-way inversion on the quotes production actually selects — 109 OTM puts and calls of the same expiry, after the tick-noise floor and the $Z_{\max}$ wing filter of section 2 — the de-Americanization moves the implied vol by a median $|\cdot|$ of 4.2 vol bps, maximum 60 (selected OTM puts: median 10 bp, max 60; selected OTM calls: median -2.1 bp, max 1.9). This is the honest statement of what de-Americanization is worth on this liquid, moderate-carry ETF chain at six months: a few vol bps in the body, tens in the put wing — material against the fitter's sub-vol-bp error budgets, an order of magnitude below the stress exhibit. (The small *negative* values on the calls are the 192-step tree's discretization against analytic Black — the finite-tree line of the error budget — not a negative premium.) Single names with hard dividends and higher rates sit between the two panels.

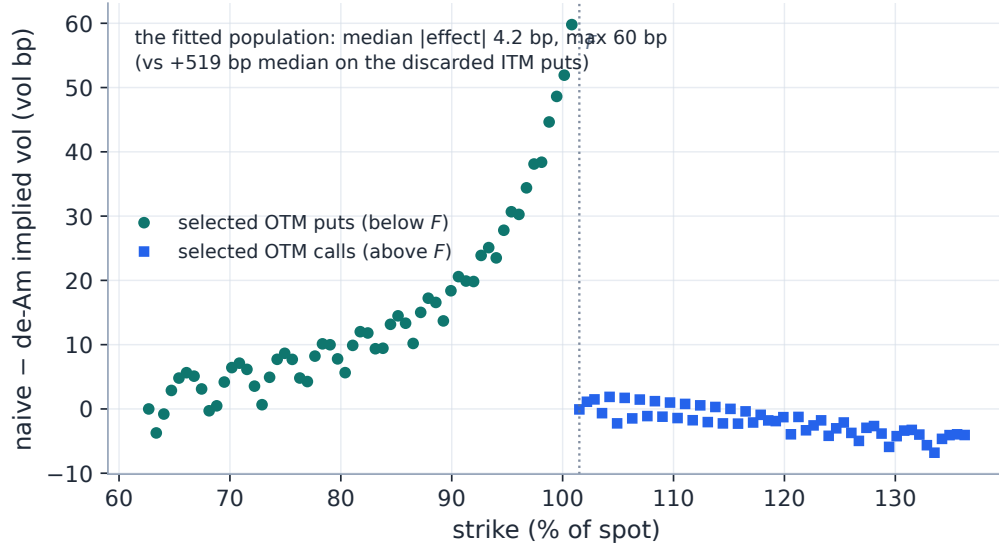


Figure 3: The same expiry, restricted to the quotes production actually fits (OTM puts below F , OTM calls above): the de-Am IV effect is a median $|\cdot|$ of 4.2 vol bps, max 60 in the put wing — the fit’s real de-Am exposure, an order of magnitude below the discarded-ITM stress exhibit. Generated by `figures/gen_deam_real.py`.

One honesty note on inputs: the delayed-tier capture’s parity-implied carry is unphysical (Note 06 documents how such feeds synthesize chains from per-contract IVs at $F = S$, $D = 1$, destroying parity information), so the carry here is supplied explicitly — $r = 4.3\%$ money-market, $q = 1.2\%$ SPY dividend yield (mid-2026) — a live instance of the caution below.

Caution

What the model does and does not cover. The tree supports a deterministic scheduled cash-dividend path (escrowed, forward-consistent via eq. (5)) *or* an equivalent continuous-carry fallback, with a CRR diffusion between ex-dates. It does not model stochastic jumps, uncertain dividend timing or amounts, stochastic rates, or stochastic borrow. And σ^* is only as good as the resolved forward and dividend inputs (Note 06): the method removes the *early-exercise* bias; it cannot repair a wrong forward.

9 What is original, and future directions

The control-variate view of tree inversion is folklore; the contributions here are the *allocation-free Numba kernel* (the per-step allocation, not the arithmetic, was the NumPy bottleneck), the *prepared-quote content cache* that scopes de-Am cost to genuine data changes rather than to fit iterations, the *measured* depth/bisection tuning that cut work without moving the numerical target, and the *wing-only, band-constrained convexity repair* (section 7) that removes the butterfly arbitrage the per-strike inversion leaves in the sparse wings while holding the dense ATM core exactly fixed. The natural next step — documented but not yet shipped — is to seed the bracket with a fast analytic American approximation (Barone-Adesi–Whaley [3] or Bjerksund–Stensland [4]) so each tree starts near its root, composing with the kernel for a further constant-factor win; these are *seeds*, not the answer, so they change no contract.

A Hyperparameter atlas

Table 1: De-Americanization hyperparameters (all hidden / internal).

Constant	Default	Role
DEFAULT_STEPS	501	Scalar CRR tree depth; European leg matches Black to $\sim 10^{-5}$.
DEFAULT_BATCH_STEPS	192	Batch/kernel tree depth.
BATCH_BISECTIONS	24	Bracketed-bisection sweeps (~ 0.002 vol bp).
SIGMA_LO	10^{-4}	Lower vol bracket floor; production lifts it above the CRR drift floor, $\max(10^{-4}, 1.5 r - q \sqrt{t/N})$.
SIGMA_HI	4.0	Upper vol bracket cap.
escrow PV	—	Discrete-cash-dividend escrowed-spot lattice (forward-consistent rescale eq. (5), cap $\alpha \leq 5$).
pre-screen buffer	$1.5\times$	Wing buffer for the conservative bid/bounds pre-screen.
prepared-quote key	—	Cache key (ticker, expiry, raw-chain <code>data_version</code> , forward, discount, dividends, t , τ , as-of).
convex_deam	<code>true</code>	Enable the wing-only convex de-Am repair (section 7); American chains only, European a no-op.
Z_{core}	1.0	ATM core half-width (units of $\sqrt{w_{\text{atm}}}$) held byte-identical by the repair.
CONVEX_TOL	10^{-3}	Convexity short-circuit: a curve convex to this (normalized price) is left untouched.
Dijkstra iterations	25	Cap on the alternating projection of section 7 (approximate, not certified).

B Performance notes

1. **Numba kernel** (section 6.2). Single scalar CRR per quote, no per-step allocation, precomputed powers, `prange` over quotes; measured $86\times$ over the NumPy fallback on a 300-quote synthetic chain (table 2; $\sim 60\times$ documented on wide real chains). Both paths through the public dispatch; identical roots asserted before the number is reported. Graceful NumPy fallback.
2. **Bisection tuning**. $45 \rightarrow 24$ sweeps, < 0.01 vol bp drift, $\sim 1.8\times$ on the isolated rail.
3. **Prepared-quote content cache**. De-Am re-runs only on a genuine data change, never on fit-setting changes — free across a calibration sweep.
4. **Output-preserving pre-screen**. Drops unusable quotes before the tree, by construction not changing any surviving result.
5. **Future (not shipped)**: BAW / BS-2002 analytic bracket seeds to start each tree near its root.

Table 2: De-Americanizing a 300-quote synthetic chain (206 invertible on both paths, identical roots; 192-step trees, 24 bisections; one untimed warm-up per path, fastest of five repeats, 20 threads).

Path	time (ms)	relative
NumPy fallback (same dispatch)	998.9	1×
Numba kernel	11.56	86×

C Reference implementation: convexity building blocks

The convexity test (8) and the convex-cone half of the wing repair are compact; the listing below shows these two *building blocks*, not the shipped repair. `min_butterfly` is the divided second difference (its minimum is ≥ 0 iff the curve is convex, i.e. butterfly-free); `convex_fit` is the convex-cone projection — the ℓ^2 -closest convex curve leaving the core anchor, written as an affine part plus non-negative hinge knots, so $\delta_j \geq 0$ is exactly the convexity constraint. The production module (`calib/convex_deam.py`) composes `convex_fit` with a bid/ask clip inside the capped Dykstra alternation of section 7 and additionally handles duplicate strikes (multiple listings at one strike) and the core/wing gating — none of which is shown here. Both listings reproduce their production counterparts to 10^{-9} .

Listing 2: The no-butterfly test and the convex wing fit (distilled from `calib/convex_deam.py`).

```

1  def min_butterfly(K, C):                                     # >= 0  <=>  C convex in
    strike (no bfly)
2      a, b = K[1:-1]-K[:-2], K[2:]-K[1:-1]                 # left / right strike
    gaps
3      fly = C[:-2]*b - C[1:-1]*(a+b) + C[2:]*a # divided second
    difference
4      return (fly / (a+b)).min()
5
6  def convex_fit(u, c, slope_floor):                          # L2-closest convex curve
    , anchor c[0]
7      cols = [u] + [np.maximum(u-u[j], 0.0) for j in range(1, len(u)-1)]
8      A = np.column_stack(cols)                             # [u, (u-u_j)_+ ...] :
    affine + hinges
9      lb = np.zeros(A.shape[1]); lb[0] = slope_floor        # leaving slope
    >= core; delta_j >= 0
10     x = lsq_linear(A, c - c[0], bounds=(lb, np.inf)).x
11     return c[0] + A @ x                                     # convex by construction
    (delta_j >= 0)

```

D Traceability

Table 3: Claim $\rightarrow$ equation $\rightarrow$ code $\rightarrow$ test.

Claim	Equation	Module	Test
σ^* root = de-Americanized Black vol; static-bound / drift-floor NaN semantics	(6), (7)	volfit/core/american.py	tests/test_american.py
Spread-preserving $\widehat{EEP}$ shift of bid/mid/ask; OTM selection; parity conversion	(2), (3)	volfit/api/quotes.py	tests/test_quotes_deam.py
24-bisection drift < 0.01 vol bp vs the 45-sweep baseline	—	volfit/core/american.py	tests/test_quotes_deam.py
Escrowed cash schedule, forward-consistent rescale	(5)	volfit/data/dividends.py	tests/test_discrete_deam.py
Kernel = NumPy fallback to tree rounding (incl. cash dividends)	—	volfit/core/american_numba.py	tests/test_american_numba.py
Prepared-quote cache: invalidation table	—	volfit/api/service.py	tests/test_prepared_cache_key.py
Wing repair: ATM core byte-identical, band-stay, convexity gate; duplicate strikes do not crash the anchor slope (cert. <code>duplicate_strikes</code>)	(8), (9)	volfit/calib/convex_deam.py	tests/test_convex_deam.py (duplicate-strike lock included)
Tick floor drops few-tick OTM quotes; tick size persists with the snapshot (cert. <code>tick_noise_quotes</code>)	—	volfit/api/quotes.py	tests/test_quotes_deam.py (tick-floor lock)
Joint (r, F) American de-bias joins the ATM sides	—	volfit/data/forwards.py	tests/test_forward_debias.py

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